

Annotated Source List

Alkubaisy, M. M., et al. "AI for Neurodegenerative Experimental Models." IEEE Xplore, 2024, doi: 10.1109/ICICAT62666.2024.10923433.

Summary:

This paper postulates that AI and machine learning can fix a big problem in neurological disease research. Results from lab models, such as those done on animals or in simulations, often do not successfully translate into useful treatments for humans, especially for diseases like Parkinson's. We have a surplus in data, but it does not capture the complexity of the human brain and its related conditions, which leads to an extremely high rate of failure. The paper reviews translational research (moving from animal models to human ones), repeatability and replication of that research, as well as the best ways to improve data quality, sharing and reuse. It proposes multiple ways to approach the obstacle. Structural equation modeling uses mathematical and statistical models to link inputs with processes and outcomes. These can model uncertainty, making predictions more accurate. Supervised machine learning builds models that learn from data to fix prior mistakes and predict human clinical outcomes. There are also unsupervised machine learning models. An experimental model, reinforcement learning, or RL, outperformed machine and deep learning algorithms, suggesting that more advanced AI can improve accuracy, efficiency, and depth of insight. Some problems that are mentioned are the black box problem and data aggregation and storage, where centralized and secure databases are proposed. The goal is to find patterns and correlation that can reveal the mechanics of the disease and help predict its progression. For future direction, AI is constantly improving and approaching human-like characteristics, but it needs comprehensive datasets, from multiple places, thus resources like foreign databases must be leveraged. To conclude, while animal and other models remain essential, many drugs that have been successful in animals have not in humans, which conveys serious issues. Overall AI and ML are very promising tools.

Application to Research:

The paper again covers disease progression and prediction. It's hard to find anything on treatment and cures. I'm having doubts as to whether PD can be cured or not. I'll have to wait until my meeting with an expert before my question can be answered.

August, Elias, and Antonis Papachristodoulou. "A New Computational Tool for Establishing Model Parameters." arXiv, 7 Aug. 2025, arxiv.org/pdf/2502.06842. Accessed 6 Nov. 2025.

Summary:

The demand for neurological care is great, and the healthcare system is struggling to meet it. This article proposes chatbots, like ChatGPT, type models to help doctors care for patients more effectively. It emphasizes augmentation rather than replacement, and AI can extend clinical capabilities by gathering large swathes of information and summarizing data, supporting decision making, and providing continuous feedback leading to constant improvement. The article lays out a 6 phase roadmap, starting with high-quality data collection, followed by decision support with AI models, integrating AI into the workspace, validating—making sure AI is safe, continuous feedback and improvement, and finally ethical issues and risk management—to protect patients and providers. The reason these measures need to be taken, is that a growing elderly population leads to more dementia cases, and due to the lack of specialists there are long delays. AI can effectively solve this problem and bring care faster and more efficiently than a human can. Each part of care is improved with AI, with it replacing many functions that are usually carried out by humans. There are, of course, ethical and risk management items that need to be addressed. The main risk is patient harm, as well as liability and privacy. Again the

issue of the black box problem comes up, where AI comes up with answers out of nowhere and clinicians cannot trust it. The key, the article says, is that the six phases of the roadmap are cyclical, not linear. Each phase leads to the next and back. Each patient interaction improves future care.

Application to Research:

This source only confirms and affirms others and strengthens the argument that AI would be an extremely large net benefit.

Choudhery, Mahmoud S., et al. "Applications of Artificial Intelligence in Stem Cell Therapy." *World Journal of Stem Cells*, vol. 17, no. 8, 2025, pp. 106086, doi: 10.4252/wjsc.v17.i8.106086.

Summary:

This article covers the use of AI in stem cell therapy, a potential cure for Parkinson's. While the article does not focus on PD, it does address the potential it has in treating neurodegenerative diseases. Stem cells are cells that can self-repair and turn into various types. Because of this they are ideal candidates for tissue regeneration. This regenerative medicine can lead to the recreation of dopaminergic cells and neurons in a PD patient's brain, effectively "curing" the disease. Different subtypes of AI, those being CNNs, ANNs, and BNs, which mimic the human neural system in how it learns, almost all subtypes being types of machine learning (ML). Using AI and these learning techniques can help researchers achieve better understanding of stem cell behaviour, differentiation, predict mortality risk, classification, modeling, drug discovery and drug delivery. There are, however, ethical limitations to consider. AI has the potential to create a wide range of problems, and there is also a current lack of trust in it from professionals, the "black box problem." There is also potential for bias in datasets, which can have negative effects. There are also other limitations to consider, including production, stiff cost, complexity, and error. Additionally, further understanding of how stem-cells work is needed, as well as standards to determine reliability of data, as well as privacy concerns with data fed to AI and who is responsible for AI's inevitable errors. AI is still in a relatively primitive state, and needs to be streamlined, if it is to be used in future practice.

Application to research:

This article is the first being added to my ASL about a potential cure for Parkinson's disease, and can serve as an introduction to potential ways to hopefully eradicate the condition.

Dixit, Shriniket, et al. "Advancing Genome Editing with Artificial Intelligence: Opportunities, Challenges, and Future Directions." *Frontiers in Bioengineering and Biotechnology*, vol. 11, 2024, p. 1335901, doi: 10.3389/fbioe.2024.1335901.

Summary:

Parkinson's disease has many potential treatments that look to cure the disease permanently. This article covers how the use of artificial intelligence in gene therapy can create leaps in advancement. Gene therapy has the potential to cure Parkinson's due to its genetic rewriting capabilities, to possibly be able to repair damaged genes that cause the condition. It can address disease causing mutations, as well knock out defective genes and even insert new genes into cells. There is a need for interventions and decisions at multiple levels, which is where AI can excel. AI's already primitive use in the field has led to increased efficiency and accuracy in the predictability of GED use in T cells. There are obstacles that must be overcome when it comes to implementing AI as a staple in genome editing, starting with data collection, coming from sources such as genomic sequencing, patient records, and experimental results. The model must be trained using supervised learning, and once training is completed, optimization must be done to streamline it. Models have already started to be used and are playing crucial roles in the area. By utilizing AI's data analysis and predictive capabilities, gene editing can become more precise,

efficient, and effective. The challenges in CRISPR and gene editing are high costs and inefficiencies in processes. There is also debate for what delivery method should be used, viral carriers, liposomes or other physical methods. AI can create delivery methods to ensure precise and accurate targeting. Ethical concerns also pose a challenge, because of the requirement of patient privacy and the need to maintain transparency relating to AI. Models have also not been sufficiently trained, and reliability needs to be developed. Data collection and a centralized database for training are also important, to prevent bias from small datasets. To sum up, AI has incredible potential but loose ends still need to be tied up and kinks need to be worked out to make AI sophisticated enough to use in the field, and not just in experimentation.

Application to research:

There are different ways Parkinson's can be treated, and this source can serve as proof that gene editing is a viable option or path. But the article does not give evidence to solving the problem of neurodegenerative diseases specifically, so more evidence needs to be found.

Ellis, Lisa D. "The Benefits of the Latest AI Technologies for Patients and Clinicians." Harvard Medical School, 30 Aug. 2024, learn.hms.harvard.edu/insights/all-insights/benefits-latest-ai-technologies-patients-and-clinicians. Accessed 10 Sept. 2025.

Summary:

This article from Harvard Medical School details and lists the benefits of AI technologies for patients and clinicians. It begins with a brief introduction containing an interview with the medical director of Anaesthesia Informatics at Beth Israel Lahey Health, Samir Kendale, MD, FASA. Kendale identifies that most healthcare providers are not yet familiar with AI, and have not trained on how it works or how to use it. Other professors and professionals say when clinicians get up to speed on AI, they will be able to use these tools to acutely benefit their practices and their patients. The article presents 5 ways AI can be harnessed, providing examples for each. AI can automate tasks to free up a clinician's time to focus more on their patients, such as capturing notes and storing them in a patient's medical record, which boosts efficiency. It might help practitioners more effectively discern imaging results, such as from an EKG or CAT scan, and while it does not replace the need for experts, it is able to certainly free up their time. AI can improve health care quality, as it has the ability to analyze large datasets and accurately point out whether a patient has a high risk of developing sepsis or prevent drug addiction after its use for medical purposes. It can also analyze huge datasets very quickly and help clinicians make more timely decisions on how to carry out diagnosis or treatment, saving hours and maybe days of research. It can also aid in decision-making, like selecting the best medicine for a patient's case, and also recognize and diagnose rare diseases that a clinician might not have encountered by scanning large datasets to find similar cases. Kendale reports these are only a few of the applications AI has in the field of healthcare.

Application to Research:

This article is extremely useful and reputable because it comes from Harvard Medical School. I have researched about the professors at the school mentioned in the article and they are real people with achievements in the academic world, and I believe them to be a reputable source that can be relied on. The article specifically provides a list and examples for my topic, and I think it is a very strong source.

Ghunaim, Lama, et al. "The Future of Pediatric Care: AI and ML as Catalysts for Change in Genetic Syndrome Management." **Jordan Medical Journal**, vol. 58, no. 4, 27 Nov. 2024, doi:10.35516/jmj.v58i4.2787.

Summary:

The health sector faces many challenges, leading to complexity and inefficiency. Thus this review explores how implementing AI and machine learning into the healthcare system can make it less complex and eliminate inefficiency. Already the implementation of these tools into the field have made diagnosis and treatment more precise and accurate. It can also lead to faster diagnosis and earlier detection. Such qualities are necessary for using suitable behavioural therapies for genetic anomalies. AI has mapped the genetic landscapes of multiple conditions using neural networks, machine learning networks that work similarly to the human brain, constantly learning new information, reinforcing what it has learnt, and applying patterns it detects into its tasks. This promotes an understanding of genomics and the development of treatment. Using AI and ML in special education also leads to a better educational turnout for students with such conditions. It can mean more tailored and flexible assistance to children with unique conditions, breakthroughs in movement-hindering conditions, as well as AI learning platforms that can give lessons in a much more stable and safer way. AI can detect movement patterns, which helps doctors study the patterns and conditions of the patient and develop a cure, as well as movement-aiding technology, such as a bionic arm that guides your hand for you. With the numerous possible applications for AI, it can “Through precise data analysis and adaptive technologies, [help] healthcare providers can enhance rehabilitation effectiveness, support mobility, and improve the quality of life for individuals with genetic musculoskeletal syndromes.” There are however, social, ethical and legal implications. Despite these, however, AI and ML are undoubtedly the future and have high potential in the health sector.

Application to Research:

The review provides little pertinent information and focuses mainly on pediatrics and related genetic disorders, not neurodegenerative diseases or gene editing.

Gond, Anjali, et al. “Artificial Intelligence Approaches for Early Prediction of Parkinson’s Disease.” *Central Nervous System Agents in Medicinal Chemistry*, vol. 25, no. 1, 2025.

Summary:

This research article talks about different ways to use Artificial Intelligence to achieve an early detection of PD. It begins by covering its history and symptoms, and then talks about Machine Learning (ML) and Deep Learning (DL). Both use voice analysis and use Neural Networks, to comprehend huge datasets to find patterns. The article postulates using digital biomarkers, tech-based ways to measure biological function, like wearables, smartphones and apps to collect data on tremors, breathing at night, eye movement, speech, gait, handwriting, and olfactory tests. It also compares different Neural Network models, such as which is best for images, time series, and dataset sizes. It concludes that using AI and ML and DL is more efficient, is cheaper, and are non-invasive alternatives to conventional diagnostics. It predicts the future as having integrated multimodal AI systems and explainable outputs.

Application to Research:

The article is from a credible database, and each author and contributor is qualified in the field. It can be used to back up findings for ways to diagnose PD, which can lead to finding a treatment.

Jackson, Edmund. “How AI Can Heal Healthcare.” TEDxNashville, uploaded by TEDx Talks, 21 Oct. 2024, youtu.be/7ZsyYCZB3Nw?si=h-chwh8q6UDTuTbG. Accessed 17 Sept. 2025.

Summary:

This video is of a lecture given by the CEO of UnityAI, Edmund Jackson. He talks about how the industry of healthcare is flawed, how it is not more successful when it could be. Healthcare has never been more advanced than it is today. Doctors and physicians are better, drugs more effective, and

therapies more successful, than they have ever been, and yet, despite this, it is failing in almost every aspect, including clinical quality, patient experience, and most of all cost. Jackson compares healthcare to other industries like technology and commerce, how processes of standardization and automation can simplify them. Healthcare can never be like this, because there is no such thing as a standard human, as each of us is different. Thus the field is extremely complex. And these physicians, drugs and therapies are only adding to it and making it worse. Jackson postulates that in order to simplify healthcare in its own way, new technology is needed. That technology is AI. The newest and most current generation of AI are machines that read and write, listen and speak, as well as think, just as a human can. Jackson uses the example of OpenAI, and the technology of its Artificial Intelligence to read the entire internet and operate within a small textbox, to propose how AI can be used in healthcare. Just as OpenAI did not simplify the internet but rather wrapped it in a simple format, AI can be used to “wrap” healthcare in a simpler way. If a doctor sees 20 to 30 patients a day in his practice, and uses the rest of his time to take notes, like revenue cycle, insurance claims, billing and regulation, AI can listen while he is with patients and take notes for him. Now the same doctor might be able to see between 60 and 70 patients a day, thanks to this simple change. Just as it did with the doctor, AI can wrap an entire hospital, keeping track of patients and their conditions, suggesting the best action for their care, coordinating it, reducing the hospital’s complexity, and increasing clinicians’ quality and productivity. AI can help detect disease much better than a human can, because it cannot make human error, and save many more lives as a result. By using AI, Jackson proposes, can cure us of the ailment of complexity in healthcare.

Application to Research:

The lecture proposes many valid points, and was given by an important figure in AI and in the topic of its application in healthcare. It gives detailed examples in a concise, easy-to-understand way. Maybe this would be a good source to use for quotes? Can look at Edmund Jackson’s work and see if it helps me in any way. This is a good source.

Bahar, Leyla. “The Role of AI in Stem Cell Studies.” YouTube, uploaded by MedicsRs Global, 25 Feb. 2024, www.youtube.com/watch?v=Vgc-mG6l-To. Accessed 15 Oct. 2025..

Summary:

This video from Dr. Leyla Bahar from Mersin University in Turkey, talks about the potential role AI can play in studies of stem cells. It first defines what AI is and what they mean by it, which was the replication of intelligent behaviour typically performed by humans instead performed by machines, with the goal of mimicking human intelligence. It then recaps the history and development of AI, from the first proposal of its potential existence by Alan Turing in the the early 1900s to present day. AI serves multiple functions in healthcare. It can create rapid diagnosis and treatment recommendations from patient documentation, treatment methods for surgery and others. The way AI can help in Stem Cell Research is largely the same, being data analysis and integration, integrating large datasets from different sources, analysis of that data, and subsequent modeling and prediction of the stem cells behaviour. It can also analyze images and prediction of cell and tissue death from those images. It also helps in drug discovery, personalized treatment plans, and prediction on how stem cells develop. These developed therapies can address autoimmune disorders, orthopedic trauma, anti-aging treatment, and neurological disorders. There are however, challenges, such as as quality control in data, standardization of datasets, validation of data, and optimization, as well as ethical concerns, such as data security and privacy protection, ethical implications, and the need for human supervision in AI’s decision-making process, to prevent inevitable failures and errors. The scientific community heavily emphasizes humans’ collaboration with AI instead of replacement, and scientific validation of all AI used, as well as

continuous improvements and collaboration with human expertise with AI's capabilities to get the best results

Application to Research:

This is a unique source, which diverts from the usual articles I cite, which adds diversity to the ASL. Additionally, the source comes from an academic source and is therefore reputable.

Lindvall, Olle. "Treatment of Parkinson's Disease Using Cell Transplantation." *Philosophical Transactions: Biological Sciences*, vol. 370, no. 1680, 2015, pp. 1-7. JSTOR, <http://www.jstor.org/stable/24505278>. Accessed 15 Nov. 2025.

Summary:

Until the late 70s, it was a wide held belief that the central nervous system of the body couldn't be repaired. However, in 1979, the brain tissue that is responsible for motor systems, the dopaminergic fetal tissue, was transplanted into rats that showed symptoms of Parkinson's. Since the transplantation of the cerebral tissue was successful and improved the rat's symptoms, it has been hypothesized that the same results could happen to humans. This journal article covers the history and the possibility of cell transplantations being used to treat Parkinson's disease. Drugs can alleviate symptoms of Parkinson's disease for many years but eventually lose effectiveness and can cause side effects such dyskinesia, or involuntary movements or twitches. Attempts didn't start with fetal tissue. Cell transplants were originally taken from the patient's adrenal gland. These transplantations produced minimal benefits and significant side effects and so the efforts were discontinued. The first fetal tissue transplantation was given in 1987 and patients showed noticeable improvement, even going off dopaminergic drugs for years. However, controlled trials did not show as much benefit overall than placebo surgery. Some patients also developed graft-induced dyskinesia. A lack of surviving dopaminergic neurons, immune rejection of the transplanted cells, and patients that were more advanced in progression, were all factors that could have contributed to the failure and lack of efficacy in the procedures. The conclusion of these theories was that patients that were more advanced in their progression of Parkinson's disease would not benefit from cell transplantation. Despite cell transplantation not becoming a standardized procedure, new discoveries were made and our knowledge of the field grew. From these studies, we learned that transplanted cells lived for a long time, and that patients showed motor improvements for upto 18 years. In some of these long living grafts, it was found that deterioration of dopaminergic cells did eventually spread to the transplanted ones, but spread and growth was slow. There is currently an EU-funded TRANSEURO trial that is reevaluating tissue transplants, with better selected, younger PD patients, as well as improved tissue preparation. However because fetal tissue is limited, other sources have been researched, including mesencephalic tissue, embryonic stem cells, and pluripotent stem cells. There are many requirements that must be met to have a clinically useful cell transplantation. There are also risks associated. Potency, safety, and patient selection are the most important loose ends to tie up. The final takeaways of the article are that cell transplantation has proven to be effective, and if the kinks are hammered, it can be a viable option for treatment.

Application to Research:

This source affirms a previous source on mesenchymal stem cells. Once again cementing the fact that stem cell therapy is the treatment for PD with the most potential.

Maclagan, Laura C., et al. "Identifying Drugs with Disease-Modifying Potential in Parkinson's Disease Using Artificial Intelligence and Pharmacoepidemiology." *Pharmacoepidemiology and Drug Safety*, vol. 29, no. 8, 2020, pp. 864-872.

Summary:

This journal article talks about how AI can be used to find drugs that can change the course and progression of Parkinson's disease. The Harvard researchers who conducted the study used AI to rank a list of 620 drugs based on how well they blocked out clumps of a protein that is associated with PD, and analyze data for whether people who took the highest-ranking drugs were less likely to be diagnosed with the condition. From these findings, researchers were able to conclude which drugs were useful and effective, and which ones weren't. This is incredible because it is an extremely fast check of real world "old" drugs that could be given a new purpose. Why Parkinson's needs drugs that only modify the disease is because current treatments only replace dopamine and help symptoms. They don't achieve prevention. And stopping these PD-associated proteins, called aSyns, could lead to slowing or even preventing the disease. There are multiple strengths as well as challenges and limitations to this drug discovery method. AI is an efficient funnel that narrows the field a lot, there is strong evidence to its effectiveness, and AI's accuracy builds trust and credibility. However, evidence and claims are not the same as clinical diagnosis. The study was only conducted on older citizens, so it's possible the drugs won't work on younger people, and the model only used data to create the drug ranking upto 2016. The article concludes that next courses of action should be to try to replicate the same results of the study they did in other regions, test whether the drugs slowed PD progression, not just incidence, re-do experiments, and if they are successful, begin clinical trials.

Application to Research:

The summary of this source concludes the research on three experimental treatments for Parkinson's, and AI's role in each of them. If just three more sources are found, maybe more, I will have built up a strong ASL.

Martinez-Martin, Nicole, et al. "Ethical Issues in Using Ambient Intelligence in Health-Care Settings." *The Lancet Digital Health*, vol. 3, no. 2, 2021, pp. e115-e123.

Summary:

This study by multiple authors including Stanford's Fei Fei Li studies the ethical implications of using ambient intelligence in a healthcare work environment. Ambient intelligence is the use of multiple devices and machine learning to monitor people and environments in clinical home settings. This form of keeping tabs can track patient condition progression, activities and care quality. However, it raises ethical, legal and social challenges – particularly around privacy, consent, and bias. These challenges show up at every stage of algorithm development, from data collection and training, up to deployment and implementation. Responsible integration of ambient intelligence requires responsible handling of personal data, checks for bias, and awareness. Ambient intelligence can track patient movement, the skill of the clinician, and reduces the burden of documentation. Outside hospitals, ambient intelligence can monitor senior citizens with chronic diseases, and identify pain. Ethical issues include privacy – the collection, storage, and use of people's personal data, bodily privacy, as in things relating to the body, such as body. Legal frameworks also hold ambient intelligence back, such as health data protection, and the right to privacy. In order to solve these issues, responsible data "stewardship" must be guaranteed. Secure storage and encrypted transmission, restricted access and training staff, and clear policies. Participants should also understand what ambient intelligence entails and give informed consent. Datasets should also be diverse and the AI should be trained so that it has minimal bias. Some ways to do this include supervised learning and third-party auditing, a third party testing and also training the AI so that bias is mitigated. As for social consequences, relationships will shift from between the patient and the clinician, to the AI being a sort of middle man, creating an over-reliance and unadvisable trust in the algorithm. There could also be distrust for the documentation that AI creates, as the algorithm itself wouldn't be trusted (Black-box problem). In conclusion, ethical concerns are not peripheral, they are not

side pieces. They are critical to the system. Privacy and data concerns are not the only ones. Bodily privacy also matters. Bias can arise at every stage, so diligence is imperative. Long-term acceptance of ambient intelligence and AI requires trust, transparency, and responsibility.

Application to Research:

This study addresses many of the ethical problems I've seen brought up through my research, and I think it can provide a way to solve these challenges. Fei-Fei Li, one of the experts who were responsible for the creation of ChatGPT, is one of the co-authors of this paper, which I think creates credibility for the paper, given the author's accomplishments and undeniable resume.

Matheny, Michael E., et al., editors. Artificial Intelligence in Health Care: The Hope, the Hype, the Promise, the Peril. National Academies Press, 2022.

Summary:

This paper covers what challenges arise in implementing AI in healthcare and how to solve them. It contains comprehensive information on how these different challenges must be addressed. AI must be trained on data that represents the entire population, by feeding it data that covers a large range of people, by putting that data on one central database to feed AI unbiased learning. By doing this, the conventional "Quadruple aim," those being population health, care team well being, patient experience, and reducing cost, becomes a "quintuple aim" adding equity and inclusion. There should also be transparency surrounding AI and what using it entails to the public, to build trust in these new digital tools. However, developers, executors, and consumers should decide what level of transparency is required for different cases. The paper also stresses that short-term focus should be augmenting clinicians and professionals' current intelligence, rather than taking over and making the process fully automated, by handling other tasks so that specialists can utilize their full capabilities as a professional. In order to maximize the benefits AI can provide, adequate training should be provided to clinicians to bring expertise of it into the field, focusing on how to appropriately use data science, as well as how to use AI services. The challenges in making the use of AI viable are many, and so we must use the methods we already know work in terms of ethical inclusivity, software development, implementation science, and human-computer interaction, to create a framework for the use of AI; in short using our current knowledge to make use of new technologies. Ultimately, AI is ready to make huge strides in health care, but we should address the challenges and explore the best solutions to put it to better use, using it in the way it can be most effective to achieve the quintuple aim.

Application to Research:

This paper is peer reviewed and each of the editors is extremely qualified and reputable in the field. The national academy of medicine is also highly reputable. This is a good source to use in the ASL.

Patel, Minesh. "A Review on Importance of Artificial Intelligence in Parkinson's Disease & Its Future Outcomes for Parkinson's Disease." International Journal of Advanced Research, vol. 9, no.9, 2021, pp. 142-147, https://www.journaliar.com/uploads/2021/09/614ad9049f9f2_IJJAR-371140.pdf. Accessed 8 Oct. 2025.

Summary:

This research article begins by reviewing what Parkinson's Disease, or PD, is and what its symptoms entail. Since PD is different for everyone, conventional clinical trials aren't viable, and therefore the use of AI has been proposed, to create specific diagnosis for everyone. AI can use machine learning and neural networks with deep learning to use different modes of data such as handwriting,

speech, gait, images and blood. It is theorized that since different people have different progressions of PD, there are distinct subtypes of the condition that AI can detect by grouping people with similar symptoms to find shared characteristics, by looking at motor data being tremors, non-motor data such as sleep and mood, medication response, and progression rate. Groups such as tremor-dominant and rapidly progressing versus slowly progressing types have already been identified. AI's learning techniques mimic a human brain, one that is everlasting and ever learning, creating, practically, a machine with a mind. A setback, however, is the lack of, despite the need for, large and diverse datasets. Generalization can have consequences and can fail across demographics, and biased training data will create inequalities. The article proposes to combat these obstacles by creating large, data-driven datasets and incorporate real world clinical data like MRI scans, which can expand diversity. In conclusion, AI has potential, but setbacks need to be addressed.

Application to Research:

There is no obvious con to including this source. It is the first one I have come across that mentions subtypes of PD. Perhaps I can send this paper in a direction that not only shows how to diagnose PD, but how to find its subtypes as well?

Mukherjee, Sayali, et al. "Recent Trends in Stem Cell-Based Therapies and Applications of Artificial Intelligence in Regenerative Medicine." *World Journal of Stem Cells*, vol. 13, no. 6, 2021, pp. 521-541, doi: 10.4252/wjsc.v13.i6.521..v13.i6.521

Summary:

Stem cells are cells that can self-repair and differentiate into different types of cells. This paper finds recent trends in stem cell research and artificial intelligence's applications in this type of regenerative medicine. Different cells are categorized into totipotent, pluripotent, multipotent, or unipotent, all based on how much of an organism they can form, with totipotent cells being able to form an entire organism, pluripotent cells can differentiate into all types of cells but not the whole organism, multipotent cells, only a definite group of cells, and unipotent only a single type of cell. Stem cells are also differentiated into adult and embryonic stem cells. Adult stem cells are preferred over embryonic ones, despite their reduced potency, because of ethical and safety-related issues. Other types of cells with different categorization methods are also covered. The paper however focuses on MSCs, multipotent stem cells that can be taken from various places, and can turn into bone cartilage, fat and muscle. They have effects on the immune system, and so are used for traditional diseases. They can also reduce inflammatory response. However, there are differences in MSC potency and functionality, depending on their sourcing, method of harvest, and delivery, with sourcing from adipose tissue being the most potent. There are also obstacles in scaling up and standardization. AI can help with this. Since AI can process swathes of data, it can immediately detect good versus bad colonies, classify cells, predict how well a patient has the likelihood of responding well to stem cell therapy, and help design tests and in clinic trials. Using deep learning methods, the study detects around a 99% accuracy. The paper concludes that regenerative medicine has immense potential but is held back, and that AI can help get rid of these obstacles. However we still need better datasets, better standards, and more study.

Application to research:

This paper focuses mostly on MSCs, which are not in the brain, and therefore this information doesn't really help in proving how stem cell therapy can help with Parkinson's, but the AI text is very useful, and postulates a good argument

North, Madeleine. "7 Ways AI is Transforming Healthcare." World Economic Forum, 13 Aug. 2025, www.weforum.org/stories/2025/08/ai-transforming-global-health. Accessed 23 Sept. 2025. North, Madeleine

Summary:

This article from the World Economic Forum lists 7 ways Artificial Intelligence is making an impact in the field of healthcare. It states that 4.5 billion people lack access to essential, base level healthcare, and the AI is the key to lowering that number, and that it is already helping doctors, but that healthcare is below average in its endorsement of it. And with the expectation of a health worker shortage within the next five years, AI can reform the field on an incredible scale, and bring universal health coverage within the very same five years. AI can interpret brain scans almost twice as accurately as a human can, and in a much smaller time frame, spot more bone fractures where humans usually err, and letting AI do an initial scan before technicians and professionals could avoid missed details and unnecessary x-rays. It can correctly predict when a patient needs to be transferred to a hospital and requires an ambulance, saving space in hospitals, and detect early signs of thousands of diseases before even the patient is aware of anything wrong. This solves the almost age-old problem of doctors identifying disease too late and losing an unnecessary life. Chatbots can also guide healthcare decisions, essentially becoming an ever-present admin, which poses huge increases in efficiency in healthcare centers. The sheer potential AI offers can lead to worries about healthcare being non-existent in the near future if it is utilized to its full.

Application to Research:

The article is very reputable, as it comes from the World Economic Forum which is a very reputable organization. Its references are also credible because they come from the United Nations, specifically the World Health Organization. It was also released very recently, which only adds to the content's credibility. Some issues however, are that the author doesn't seem to have any qualifications and has written many different articles for the WEF, and one paragraph is repeated word for word. It's also a .org, so it's not the best source in that sense. But overall its quality is good.

Saxena, Avinash. Personal interview. 18 Jan. 2026.

Summary:

This interview addresses many issues that concern a wide implementation of AI. The black-box problem is formally defined as an AI's inability to show why it came to a decision and its reasoning behind it. The main way to fix this problem is to have a human in the loop, monitoring the AI and making sure it doesn't hallucinate, at least for the foreseeable future, because AI isn't capable of advanced processes yet. Use red teaming, a process in which, over a period of a few months, the AI is given all kinds of "Doomsday" scenarios to see how it responds, and whether that response is what the team wants and expects. Using guardrails and interpretable models also helps. To prevent people's personal data from being used, again using guardrails and redacting unnecessary information is key, to treat the AI as any employee, having regular audits and checks. Training AI is expensive, and to minimize expenses, using models with a cache that stores information, or even using your own, self-made model are the most common methods. AI is purely meant to augment rather than replace humans. Neural networks are of different types. Some have a memory that influences their future decisions, and some only do one-time, linear calculations. The use case matters immensely when trying to decide which to use, and one should always choose the simplest model that is most valuable, and then increase complexity as needed.

Application to Research:

The interviewee works specifically on AI.

“What Role Does AI Play in Developing Gene Therapies?” Patsnap.com, 2025,
synapse.patsnap.com/article/what-role-does-ai-play-in-developing-gene-therapies.

Summary:

Gene therapy is a procedure that consists of the changing and modification of the genetic material to treat disease. It can include the introduction, removal, or correction of genes. At the most basic it is to reprogram cellular function. Currently there are gene therapy drugs on the market that release genes and proteins to cure conditions and ailments. This article covers how Artificial intelligence (AI) has changed the landscape of data interpretation immensely. They are used to identify patterns in data that are used to further understanding, predict outcomes of genome editing, and optimize design for systems. To sum up, the implementation of AI can reduce burden, streamline testing and simulation, and accelerate the discovery of important therapeutic strategies. AI models learn like humans, and they can improve as new data is fed to them. Currently AI has been integrated into genome editing technologies, which have displayed these results. As of current, AI has displayed rapid data analysis, enhanced nucleic acid design, predictive capabilities, and has streamlined R&D and reduced costs. Multiple studies have been done that show that employing deep learning and machine learning algorithms has led to large improvement in editing outcomes. Despite AI's benefits, there are ethical challenges involved. Large, diverse, comprehensive datasets are needed to make the algorithm reliable. AI needs to be trusted, and as it can seemingly generate answers out of nowhere, clinicians find it difficult to associate it with reliability, leading to the creation of a “black-box” problem. There are also issues raised regarding misuse and data and privacy violations. Emerging trends do show that AI systems have started to get integrated, and explainable AI systems have become more prevalent, encouraging transparency in the algorithm's decision making process. Future directions include research into longitudinal data collection, collaborations between computer scientists, biologists, and clinicians, standardization between AI models, as well as ethical research. Artificial intelligence plays a large role in gene therapy by augmenting every stage of the process. However, there are issues that need to be addressed. However solutions to these problems are on the horizon, and by conducting research into these issues, AI can become a permanent and essential part of the field.

Application to Research:

This source provides a lot of useful information, which is consistent with other scholarly articles I have reviewed. However, its website domain, lack of an author, and ads are not professional and therefore make this article not as credible.

Wu, Peng, et al. "The advantages of artificial intelligence-based gait assessment in detecting, predicting, and managing Parkinson's disease. Edited by Oscar Arias-Carrión. *Frontiers*, 11 July 2023, www.frontiersin.org/journals/aging-neuroscience/articles/10.3389/fnagi.2023.1191378/full. Accessed 2 Oct. 2025

Summary:

This journal article covers the use of AI-enabled gait analysis and diagnosing Parkinson's disease, or PD. Currently, doctors rely on clinical methods, identifying obvious and explicit symptoms, at around an 80% success rate. However, by the time PD is diagnosed, about 60% of neurons that PD affects are lost. Early detection is crucial yet difficult and rare. Drugs like levodopa alleviate symptoms but are not a definite cure, and using drugs over a long period of time can lead to side effects like dyskinesia. Currently the best approaches to treating PD are medicine, exercise, therapy and emotional support. AI can be a positive addition to treatment because it can find patterns in large datasets and detect PD early, by analyzing gait through devices and sensors, and accurately predict Freezing of Gait,

or FOG. AI can detect minute details such as step length, cadence and acceleration. AI can also detect symptoms of PD in handwriting and speech, two of the earliest signs, and is extremely accurate. It analyzes the period before FOG, called pre-FOG, around 1-6 seconds prior. The article also highlights the benefits of exercise and movement in preventing PD and boosting brain health, using treadmills as well as robotic training. Multiple devices can be used for gait analysis, including smartphones. Some challenges with using AI are the black box problem, the fact that AI makes predictions without explaining why, which can decrease trust with professionals, the difference between gaits of patients when they are on medication versus off, small datasets and bias in datasets, as well as pre-FOG time windows. For future direction, transfer learning, semi-supervised learning are good, though not as good as person-specific, supervised learning. Standardized datasets and clinically explainable AI are also needed.

Application to Research:

This article is peer reviewed and was published recently, only two years ago. It is a credible source and will be used.

Yang, Ni, et al. "Motor Symptoms of Parkinson's Disease: Critical Markers for Early AI-Assisted Diagnosis." Edited by Liane Kaufmann, Frontiers, vol. 1, 2025, www.frontiersin.org/journals/aging-neuroscience/articles/10.3389/fnagi.2025.1602426/full. Accessed 7 Oct. 2025.

Summary:

This journal article covers the markers associated with the symptoms of Parkinson's Disease (PD), along with the potential of AI in those specific areas. It defines the terms AI, Machine Learning and Deep Learning, and how they can help in identifying the motor symptoms of PD, including tremors and other signs. Patients have been observed to have difficulty in controlling their eyes, focusing on and tracking a certain object. AI can isolate the slightest abnormal movement in the eyes. A model can be created from the data to successfully diagnose PD. Beyond simply detecting symptoms, AI can also figure out the severity of the disease, which is much better than a human ever could. The article addresses the challenges of using these different methods, and potential future directions. In the case of eye movement, head movement or head tremors can severely impact data. PD patients also tend to have more relaxed, and less control of their facial muscles as a whole. AI can detect the changes in facial expression to diagnose PD. Data can be taken of a patient before getting Parkinson's to compare to the current version, which can be used to accurately detect the condition. Apps have been developed to help detect PD remotely. However, AI needs large datasets to be effective, and in the case of facial expressions, datasets tend to be small. The expression of emotions can also imitate the progression of the disease, which can lead to inaccurate diagnosis. AI needs to be more effectively trained with nuance to be effective. Parkinson's can also affect the voice of a patient, and even their sentence formation and grammar. AI can detect these niche changes and classify them as PD, and has multiple ways of identifying it, those being speech analysis, prosody, and cognitive-linguistic. These methods have already started to be widely implemented. The deterioration of a person's handwriting and tapping skills can also be analyzed to detect the condition. These AI models have been used in smart pens and tablets to collect data. The use of sensors has also been studied. Obviously, there are complications due to the fact that a patient's emotional state can affect how they write, but models are still highly accurate. The use of AI models to diagnose Parkinson's is extremely mature, however some kinks still need to be worked out. In conclusion more experimentation and study has to be done to achieve a satisfactory condition of AI models

Application to research:

The article has many many articles cited and its obviously gonna be one of the most valuable sources I use, but it doesn't have anything related to treatment or cure, so it doesn't give me what I lack.

Youngmann, Brit, et al. "A Machine Learning Algorithm Successfully Screens for Parkinson's in Web Users." *Annals of Clinical and Translational Neurology*, vol. 6, no. 12, 2019, pp. 2503-2509, doi: 10.1002/acn3.50945.

Summary:

This study investigates into how AI can detect and diagnose Parkinson's by analyzing how people interact with a search engine, without using what they actually type. It analyzes behaviours such as mouse movements and typing, as well as timing and other interactions. This can be important because many people with an early stage of Parkinson's don't seek medical care, and thus diagnosis is often late. Interactions with web applications may show motor changes caused by the condition. Since it is easy to conduct tests via search engine, a possibility for screening on a large scale opens. The primary goal of the study was to determine whether a machine learning model can diagnose PD remotely, through the use of web interaction analysis. It fulfills a lot of ethical criteria that other ways of diagnosis do not. Data collected was fully anonymous, it operated under the browser's terms of service, there was no informed consent of the data collection required, no identification of individuals, and the study was approved by an ethics committee. However an ethical issue that was raised was the fact that people were getting diagnosed without their permission. The study used Bing, a Microsoft search engine to conduct the study, and collected data on cursor movement, clicks, keyboard behaviour, and the number of searches a person conducted. Multiple study "groups" were created, the first being people that self-identified as having PD, the second being people talking about someone else having PD, and a portion of the general population that had nothing to do with the condition. A fourth, unclassified group, users searching for Parkinson's related info, and a fifth, users reporting early symptoms or family history with PD, was also used. The first two groups were used to train the AI model, and the third, fourth, and fifth groups were the ones that were actually tested. A key aspect of this study that makes it unique is that it used a Long Short-Term Memory (LSTM) network, a neural network that learns patterns from time-series data, which means they have a memory that can remember previous data, something other networks cannot do. The model was built to return a likelihood score on a scale of 1 to 5. It would analyze multiple sessions of a user, and if the majority of the movements during that session raised a red flag, then it classified that patient as having PD. The results of the study showed a high accuracy. It found results in the general population that were comparable to common medical screenings. It also analyzed PD likelihood scores over time, and tracked the progression of the disease, and users that the model classified as PD showed a similar progression to known PD patients, signalling correlation and consistency. It has many strengths, including that it has a large real-world dataset and that it is fully validated across multiple populations. Its limitations do include that it cannot tell PD from other conditions with similar symptoms, its self diagnosis was not verified, there was no physical examination, and ethical issues that include unsolicited and uninformed diagnosis. A key conclusion is that ethical responsibilities now extend to technology companies from clinicians when you implement AI to address medical conditions.

Application to Research:

This source is unique because it has done something I have never found before. It includes a strong neural network that overcomes the limitations others have, making it have more potential to find a treatment for Parkinson's.